

AYMANE AIT DADS

Applied AI Engineering Intern | LLM Fine-Tuning · Anomaly Detection · ML Pipeline Development

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PROFESSIONAL SUMMARY

Fine-tuned a 30B Mamba-Transformer MoE model with LoRA on 7,828 chain-of-thought reasoning examples, reaching top 10% public leaderboard (0.80/1.0) in the \$106K+ NVIDIA Nemotron Kaggle competition - directly demonstrating LLM pipeline design and ablation-driven benchmarking across model configurations. Built a clustering and classification pipeline on 100K+ fiber-optic network records using Random Forest and K-Means, reducing manual analysis time by 60% and delivering structured outputs to a network optimization team. Proficient in Python, PyTorch, Scikit-learn, Pandas, Hugging Face Transformers, anomaly detection, and LLM orchestration with tool-calling and RAG. Available for a Summer 2025 internship; currently based in France and open to USA/Canada relocation (visa sponsorship required).

CORE COMPETENCIES

LLM Pipeline Design | Anomaly Detection | Clustering & Root Cause Analysis | Python & PyTorch | ML Model Benchmarking | Unstructured Data Processing | Cross-functional Collaboration | Data Analysis & Visualization

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built a clustering pipeline using K-Means to map customer performance profiles to **5 commercial service tiers** across thousands of fiber-optic network nodes, output adopted directly by the network optimization team.
- Developed Random Forest classification models on **100K+ fiber-optic network records** (upload speed, latency, jitter) to diagnose and classify network quality degradation across the national infrastructure.
- Reduced manual analysis time by **~60%** through automated feature engineering and model evaluation pipelines built with Python, Scikit-learn, and Pandas.
- Delivered stakeholder presentations translating ML model outputs into actionable maintenance recommendations, bridging **data science and operations teams** using Power BI dashboards.

PROJECTS

NVIDIA Nemotron Reasoning Challenge

Kaggle Competition · In Progress (June 2026) · Top 10% Public Leaderboard

- Fine-tuned a **30B Mamba-Transformer MoE model** (3.5B active params via MoE routing) using LoRA (r=32, ~888M trainable parameters) on 7,828 chain-of-thought examples, reaching 0.80/1.0 vs. 0.49 base score in the \$106K+ prize pool competition.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** in strict sequence - including monkey-patching allocator warmup, stubbing Mamba/Cutlass imports, and replacing Triton kernels with PyTorch fallbacks - and conducted 20+ single-variable ablations confirming completion-only loss and full-linear LoRA as critical design choices.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton

Anomalous Sound Detection - Industrial Equipment

EURECOM · 2025 · AUC 0.80 Predictive Maintenance

- Designed an unsupervised anomaly detection system for slide-rail manufacturing equipment using a Transformer autoencoder on Mel spectrograms with SpecAugment, achieving **AUC 0.80** with zero labeled anomaly examples.
- Built an end-to-end signal processing and feature extraction pipeline mapping raw audio to structured fault signatures, directly matching the JD requirement to **cluster failure signatures and surface probable root causes** for engineering triage.

PyTorch · Mel Spectrograms · SpecAugment · Transformer Autoencoder

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Statistics, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

LLM Fine-Tuning & NLP: LoRA / QLoRA | SFTTrainer | Hugging Face Transformers | vLLM | RAG | LangChain | Chain-of-Thought

ML / Data Science: PyTorch | Scikit-learn | Pandas | NumPy | Anomaly Detection | Clustering | Ensemble Methods

MLOps & Infrastructure: Python | SQL | Git | Docker | Supabase | Power BI | CUDA / Triton Debugging